

Chapter 81

Paint

This chapter describes how to use Fusion's non-destructive Paint tool to repair images, remove objects, and add creative elements.

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Paint Overview

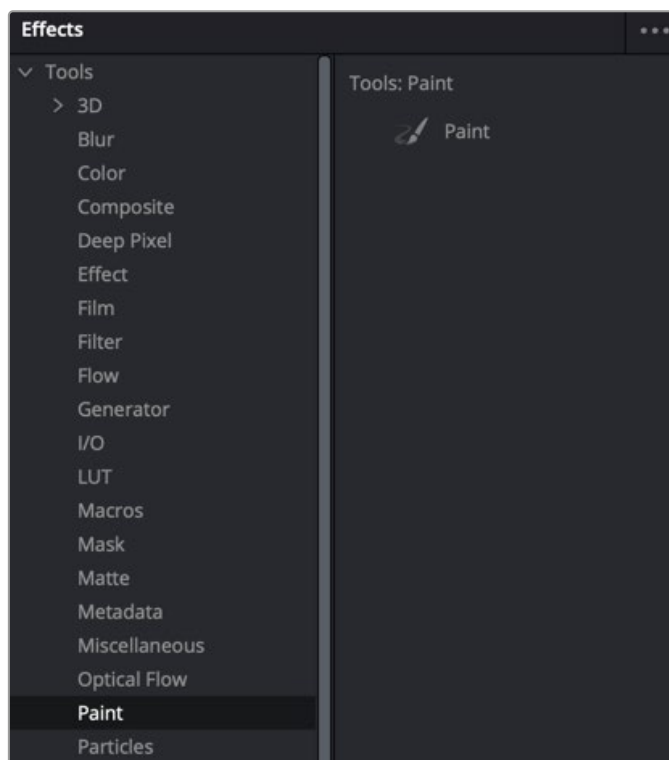
The Paint node is a procedural paint tool, which means that each paint stroke is a live, editable object that's drawn with properties that you can mix and match to address a wide variety of painting tasks. You can use it to paint masks, retouch images, perform beauty work, clone out objects, or even create motion graphics. Each element of a paint stroke can be altered long after you apply it. Since strokes are editable, you can apply, change, ignore, delete, and even reorder them in a node tree.

Types of Paint Nodes

There are two types of paint nodes in Fusion. The Paint node is a full-featured creative and retouch vector-based paint tool that requires an input on which to paint. The Mask Paint node lets you specifically paint an alpha channel to limit the area of an effect. It allows you to create paint strokes on an alpha channel without needing to have an input.

- The Paint node is located in the Paint category of the Effects Library.
- The Mask Paint node is located in the Mask category of the Effects Library.

The main difference between these two Paint tools is that the Mask Paint tool only paints on the Alpha channel, so there are no channel selector buttons. The Paint tool can paint on any or all channels. The majority of this chapter covers the Paint node, since it shares identical parameters and settings with the Mask Paint node.



The Paint node available in the Paint category of the Effects Library

Setting Up the Paint Node

The Paint node has two inputs. Typical of most Fusion nodes, the orange input background is the primary input for connecting the “canvas” or image to paint on, while the second blue input is an Effect Mask. Unlike the Mask Paint node, the Paint node requires a background input to begin painting.

Setting the Paint Node’s Resolution

No matter how you use the Paint node in your node tree, the Paint node assumes the resolution of the background input image as your working resolution for that operation. Although the Paint tool is in fact resolution independent and you can change this assigned resolution at any time, it’s essential to make sure you properly set the resolution of the media you’re working with, because it will affect operations such as motion tracking that you’ll want to use as part of your paint workflow.

Because of this, the Paint node requires a background input to set the resolution of the “canvas” you’ll be painting upon. To do this, you can set up a Paint node in the node tree in one of two ways: painting directly on an image or using Paint as the foreground.

Painting Directly on an Image

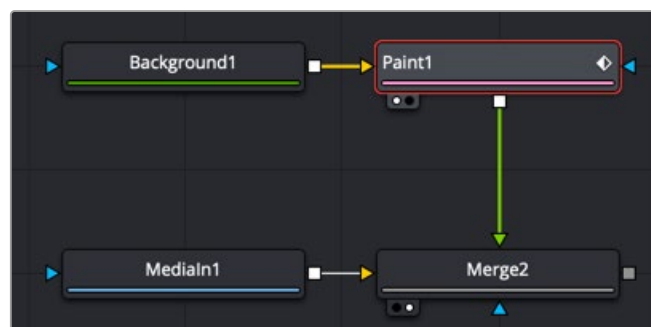
You can connect the image you want to paint on directly to the background input of the Paint node. This is the easiest and cleanest node tree construction, but it doesn’t provide much in terms of compositing flexibility.



The Paint node is inserted directly after the node it is painting on

Using Paint as the Foreground in a Merge Composite

An alternative setup is to use a Background node to set the resolution for the Paint node, compositing the result over the actual background using a Merge node. Working this way lets you use the Merge node’s Apply Mode setting (also referred to as composite modes) to control how your paint strokes are composited against the image, but it does require a bit more setup.



The Paint node is composited over the image you want to paint on using the merge

Setting this up requires some configuration of the nodes. The Background node must be fully transparent and, unless you are doing something simple like using the Stroke tool set to Color to paint over an image, you must drag the image you want to clone or smudge into the Source Tool field in the Paint node's inspector. These steps are described in more detail later in this chapter.

Paint Node Workflow

You begin painting by first selecting the paint stroke type from the Paint toolbar above the viewer. There are ten stroke types to choose from as well as two additional tools for selecting and grouping paint strokes.

The stroke types and tools are described in detail, see *Chapter 113, "Paint Node,"* in the DaVinci Resolve Reference Manual or Chapter 53 in the Fusion Reference Manual.

The primary tool for painting and cloning is the Stroke tool. The Stroke tool is a fully animatable and editable vector-based paint stroke and initially uses a duration of the entire global range.



The Stroke tool is most often used for cloning, beauty work, and creative paint

Select the Correct Paint Stroke Type

Instead of having multiple dedicated Paint nodes for different operations, the procedural nature of Fusion's Paint tool means there is just one tool with a variety of different stroke types. Each stroke type serves a different purpose. The Paint toolbar above the viewer allows you to choose between different stroke types and drawing tools. These are grouped into a few categories.

Single-Frame Stroke Types

The Multistroke and Clone Multistroke are explicitly designed for single-frame retouching paint jobs like removing raindrops in a shot that's supposed to be a sunny day or removing scratch marks and dust when restoring vintage content. When handling these types of jobs, these two Multistroke options are faster than the other stroke types, but they're not editable later on. This means you must set up the size and function of the brush as well as the duration for the paint stroke before you paint.

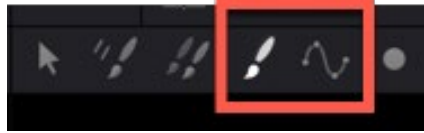
The duration of each of these stroke types is one frame by default, but this can be changed using the Stroke Duration slider in the Inspector. Multistroke and Clone Multistroke are basically the same tools, except Clone Multistroke automatically configures the tool for cloning. In contrast, the Multistroke requires you to set up the tool for cloning manually.



The Multistroke and Clone Multistroke are non-editable single frame stroke types

Editable Stroke Types

The Stroke and Polyline are similar in that they can be modified and animated at any time. Also, they both begin with a duration lasting the entire comp, but that also can be changed using the Keyframes Editor.

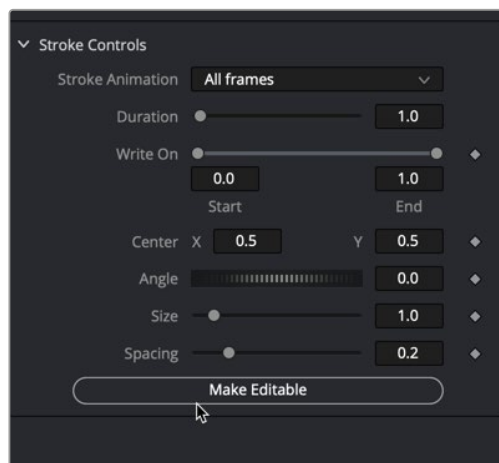


The Stroke and Polyline strokes are editable and last for the entire comp

The Stroke Tool

One of the most flexible editable stroke types you'll use for many tasks is the Stroke, because it is fully animatable and editable. You can animate all elements of the Stroke, and you can use the Write-on/Write-off parameters to control how the stroke appears onscreen. You can also connect to a tracker from the Center point of the Stroke if you want to make the stroke follow specific onscreen motion.

By default, the Stroke type does not expose control points for the shape of the path. You can move and track the center and rotation of the Stroke, but the individual control points that create the spline are hidden. To reveal the control points, you can open the Stroke Controls at the bottom of the Inspector and click the Make Editable button.



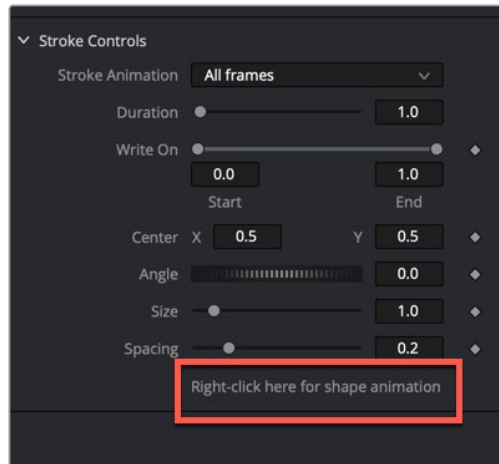
The Stroke's control points can be revealed using the Make Editable button

Although the Stroke type is the most flexible, that flexibility can come at a performance penalty if you're painting hundreds of strokes on a frame. For larger numbers of strokes that do not need to be animated, it's better to use Multistroke or Clone Multistroke, as these are more processor efficient.

The Polyline Stroke Tool

The Polyline Stroke acts more like a drawing tool than a paintbrush. It includes the same functionality as the Stroke tool, except that it is created not by dragging or "painting" like a paintbrush, but by clicking to create a spline path, as you do with masks and motion paths. Without even creating a stroke in the viewer, the Polyline Stroke can connect to existing polylines like a mask or a motion path.

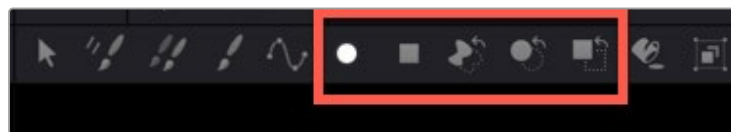
If a motion path is published, right-clicking on the Shape Animation label at the bottom of the Polyline Stroke's Stroke Controls allows you to use the Connect To menu to assume the shape of a motion path or mask. You can also use this method if you import SVG graphics and want to "paint-on" the outlines.



The Polyline Stroke Shape Animation label

Shape Drawing Tools

Five shape-based drawing tools allow you to draw shapes and either fill them with a color or clone an area from a source image. All these tools act similarly to the Stroke and Polyline stroke type in that they are editable at any time and have a default duration spanning the entire global range of the comp. However, you can edit the duration at any time in the Keyframes Editor.



The Shape Strokes are used to create shapes or clone areas based on shapes

All of the Copy [Shape Name] stroke types require that you connect the source node you are cloning from into the Paint node, and set the Fill Type menu to Image.



The Copy Shapes require a Source to be directly connected to the Paint tool and set to Image

Setting the Brush Size

After selecting the Stroke type, the Brush size can be set in the Inspector or more intuitively in the viewer. With the Paint node selected in the Node Editor and the pointer positioned over the viewer, you can see an outline of the current brush size. To change the brush size, hold down the Command key and drag. The circle changes size, so you can set it relative to other objects you may be painting over.



Brush size can be changed interactively in the viewer

Choosing an Apply Mode

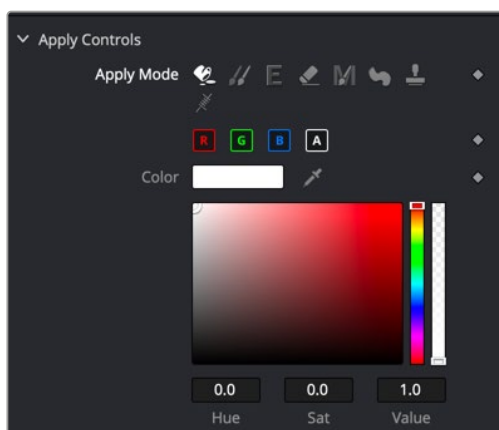
The Apply Mode buttons determine the functionality of the paintbrush. There are eight Apply modes that set the brush to do things like paint a color, clone from a source, smudge an area, or remove thin wires.



Apply Mode buttons determine the paint brush functionality

Picking a Paint Color

There are several ways to pick the paint color and opacity for a colored brush stroke. You use the Fill button in the row of Apply modes when you want to paint with a solid color.



The Color swatch shows the current color
Clicking it opens the OS Color Picker window

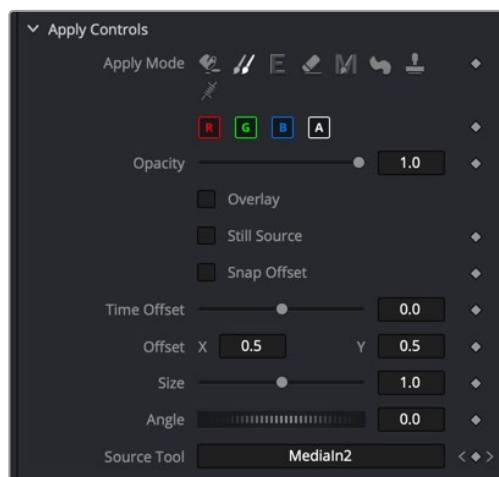
To select a color for the paint brush, do one of the following:

- Click the color swatch to open a standard OS Color Picker window.
- Drag the Eyedropper into the viewer.
- Drag inside the color chooser to select a saturation and luminance. Drag on the sidebars to change the hue and transparency.

When you paint, each stroke is unpremultiplied, so adjusting the Alpha slider in the Inspector does not affect what you apply to the RGB channels. However, changing opacity affects all four channels.

Cloning from the Frame

Choosing the Clone Apply Mode allows you to paint from one area of an image over another area. This is the most common use of the Paint tool. It allows you to remove objects or artifacts from a clip by covering them up with another area of the frame. Depending on the Stroke type chosen, you may clone on either a single frame or for the entire duration of the clip.



The Clone Apply Mode allows you to sample from one area and use it as a source to paint over another area

You can use the Clone Apply Mode to clone from the same image connected to the Paint node's background input or a different source from the node tree.

To clone from a different area of the same frame:

- 1 Select a Stroke type from the Paint toolbar above the viewer.
- 2 Using the size slider in the Brush controls section of the Inspector, set the size of the brush.
- 3 From the Apply Mode buttons, select the Clone mode.
- 4 Option-click over the area in the viewer you want to use as the source. A dot appears showing you the center of what you're sampling from.



The Clone source starting area identified by the X and the paint brush size represented by the circle

- 5 Paint over the area you want to cover up using the source pixels.

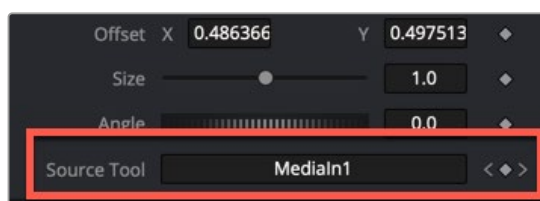


The Clone completed after selecting the source area and painting over the flag pole

When trying to erase objects or artifacts from a clip using the Clone Apply Mode, it can sometimes be easier if you sample from a different frame on the same clip. This works well when the object you are trying to clone out moves during the clip, revealing the area behind the object. Sampling from a different frame can use the revealed background by offsetting the source frame.

To clone from a different frame of the same clip:

- 1 Select a Stroke type from the Paint toolbar above the viewer.
- 2 Set the size of the brush.
- 3 From the Apply Mode buttons, select the Clone mode.
- 4 Drag the clip (Medialn1 or Loader) from the Node Editor into the Source Tool field in the Inspector.



The Paint Inspector with the Medialn1 dragged from the Node Editor into the Source Tool field

- 5 Click the Overlay checkbox to see the current frame and the offset frame superimposed.
- 6 Drag the Time Offset slider to select the source frame you want to use.
- 7 Option-click over the area in the viewer you want to use as the source or to offset the source frame's position.



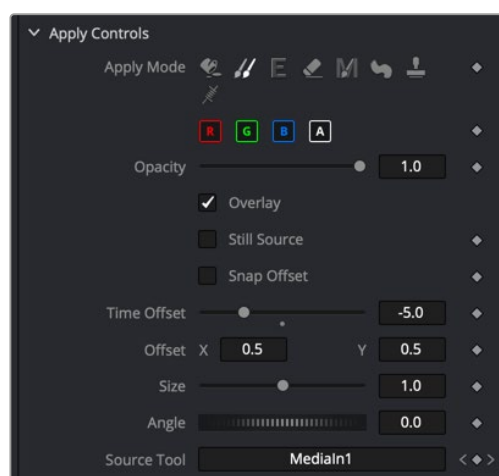
Overlay shows two frames overlapped with Time Offset, allowing you to clone from one frame onto another

- 8 Paint over the area you want to cover up using the source pixels.



The plane is half painted out using on the Overlay with Time Offset

- 9 Disable the Overlay checkbox.



The Clone Apply Mode can use a different frame from the same clip

TIP: When using a Clone Apply Mode, you can hold down the O key instead of clicking the Overlay checkbox in Inspector to see the Overlay. Releasing the O key will return to normal viewing without the Overlay.

Editing Paint Strokes

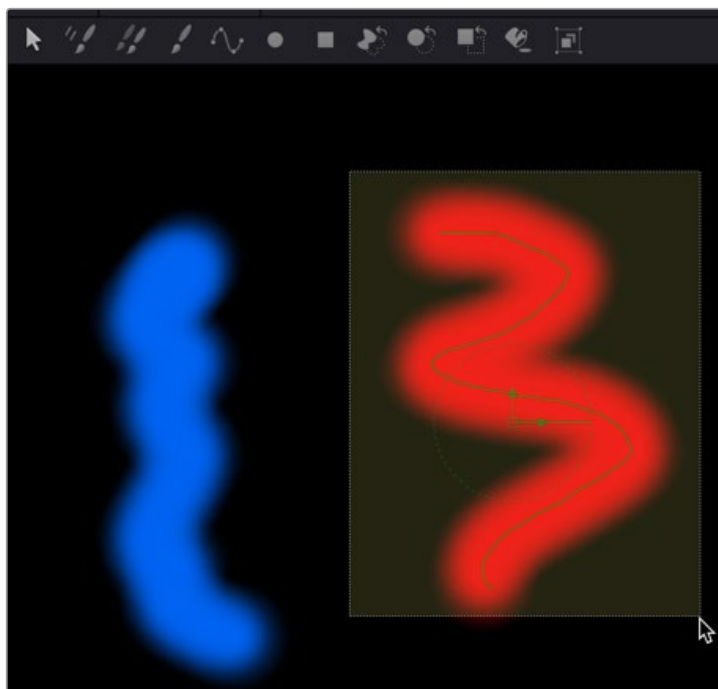
Once you've painted using the Stroke or Polyline stroke type, you can change the look of the stroke by selecting it and updating the parameters in the Inspector. Selecting the stroke requires you to switch to the selection tool in the Paint toolbar above the viewer. Using the Paint node's selection tool, you can either click once on a stroke or drag a bounding box around a stroke to select it for editing.

TIP: To select multiple strokes, you can Shift-click or Command-click to select and deselect multiple specific strokes, or you can drag a selection box around all strokes you want to select.

Although you can make changes in the Tools tab in the Inspector, the Paint node uses both the Tools tab and the Modifiers tab. In the Tools tab, you can create new brush strokes and select a stroke in the viewer to edit. The Modifiers tab presents a list of all the strokes for the selected Paint node, which makes it easy to modify any previously created paint stroke.

NOTE: Multistroke and Clone Multistroke each only appear as one item in the Modifiers tab no matter how many strokes you create using those tools. Those two tools are not editable after creating them.

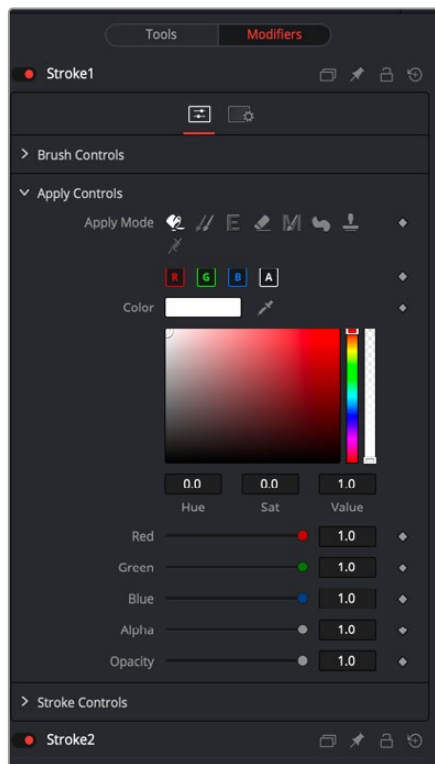
The same controls you used in the Tools tab to create the strokes are located in the Modifier's tab to modify them. You can also animate each individual stroke.



The Stroke or Polyline Stroke type can be edited by selecting the stroke in the viewer

Editing Paint Strokes in the Modifiers Tab

When you paint a stroke, the settings for that stroke appear in the Inspector's Modifiers tab. You can then change the settings in the Tools tab for the next stroke you are about to paint. Each time you click, drag, and release the pointer button, you create a new stroke. Each stroke is numbered in the Modifiers tab, where it can be selected and edited.



Once you stop painting a stroke, it's added to the Modifiers tab along with an additional Stroke modifier that represents the next stroke. For instance, if you paint your first stroke, the Modifiers tab shows your stroke as Stroke1 and then a stroke 2 as well, which represents the next stroke you create. You always have one more stroke in the Modifiers tab than strokes in the viewer.

Each stroke is listed in the Modifiers tab for editing

Deleting Strokes

There are two ways you can delete paint strokes.

To delete any individual stroke, do the following:

- 1 Select the Paint node.
- 2 Click the Modifiers tab.
- 3 Right-click over the Stroke header you want to delete and choose Delete from the menu.

To delete all paint strokes you've made on every frame, do one of the following:

- Click the reset button in the upper-right corner of the Inspector.
- Delete the Paint node in the Node Editor.

Animating and Tracking Paint Strokes

In some ways, animating paint strokes is no different than animating any other effect in the Inspector. Each parameter that can be animated includes a gray diamond Keyframe button along the right side. Clicking the Keyframe button sets a keyframe on the current frame and enables auto-keyframe mode for the parameter. However, more commonly, the paint stroke is tracked using one of Fusion's trackers, or for motion graphics, animated using the Write-On Start and End sliders.

Animating with Write-On Controls

The Stroke and Polyline stroke types include Write-On controls located in the Stroke controls section of the Inspector. These Write-On controls animate the appearance of a stroke along the path. You can animate the Write-On controls using either the Stroke Animation drop-down menu or using the Start and End sliders.

Stroke Animation Drop-Down Menu

The Stroke Animation drop-down menu includes six options for auto-animating a paint stroke. The first two options do not truly animate the stroke as much as set a duration. The Limited Duration option uses the Duration slider to set the number of frames the stroke is onscreen.

To auto-animate the stroke, you can choose one of the three Write options or the Trail option.

Choosing Write On automatically creates a write-on animation. The duration is set by two keyframes that get added when you choose Write On from the menu. The Start keyframe is set on the frame where you first created the stroke. The End keyframe is added on the current frame when you choose Write On from the menu. The remaining options in the menu set their Start and End keyframes similarly but change the direction of the animation based on the menu selection.

Write-On Start and End Parameters

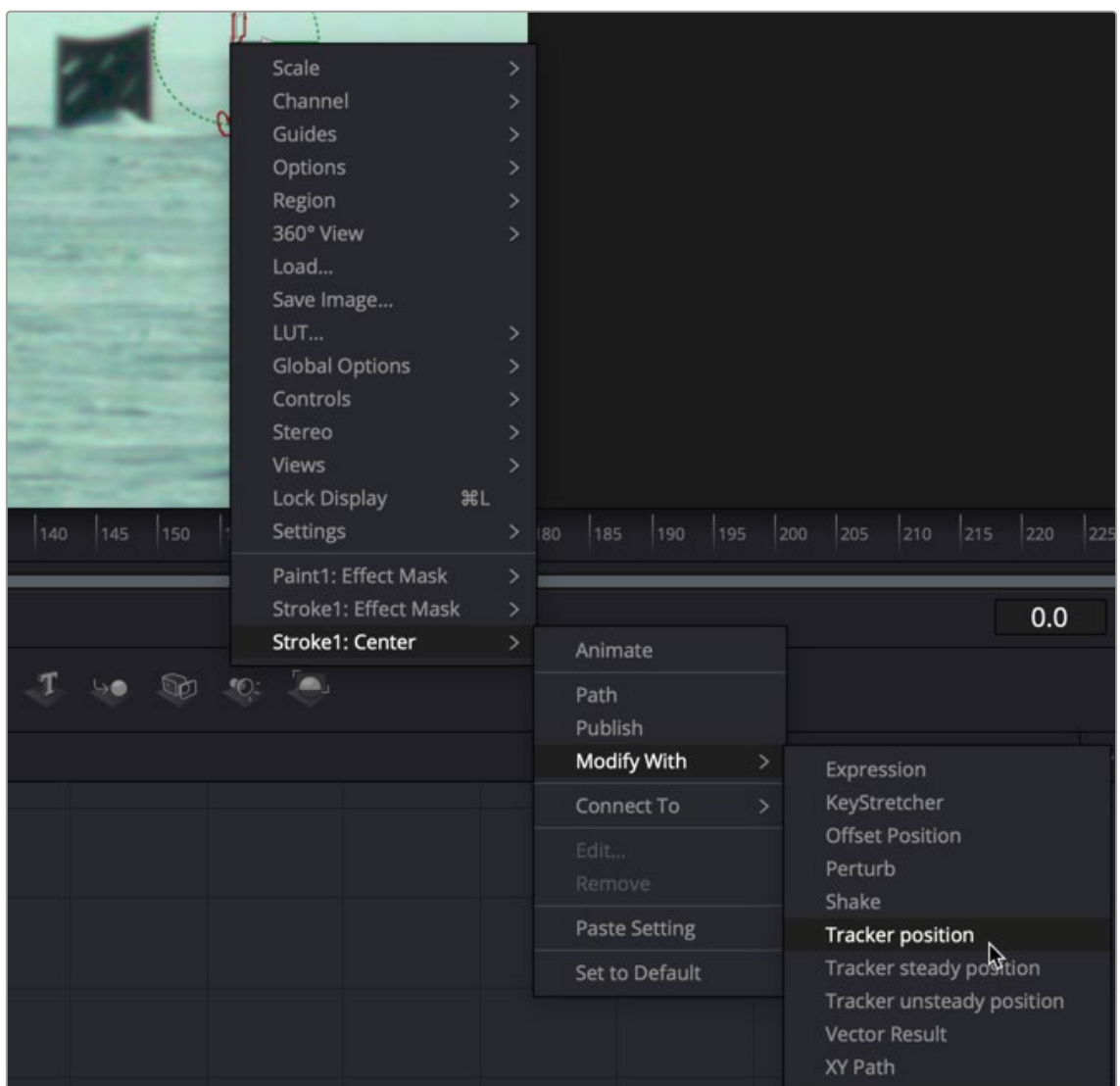
The Write-On Start and End parameters allow you to manually control the start point and end point along any stroke's path, and use keyframes to animate each parameter individually. The Start parameter determines the point at which the stroke begins, measured as a percentage offset from the beginning of the stroke's path. For example, a Start value of 50 moves the starting point of the stroke to the middle of the stroke's path. The End parameter works the same way but from the other end of the stroke. You can animate a stroke onscreen, creating a handwriting effect by setting keyframes for the End parameter from 0 to 100 over several frames.

Tracking a Paint Stroke

You can animate the position of a paint stroke using any of Fusion's trackers. For instance, if you cloned out a flag pole from a clip, but the camera moves, you can track the flag pole and attach the resulting path to the paint stroke.



Trackers can be attached to the Center parameters of a paint stroke



Right-clicking over the Stroke's Center control in the view allows you to apply a tracker modifier

To attach a tracker to a paint stroke:

- 1 With the Paint node, select the Stroke brush type and clone out an object on a frame.
- 2 In the Paint toolbar above the viewer, click the Select tool.
- 3 Drag a selection box around the stroke to select it.
- 4 Right-click the center control on the stroke, and then choose Stroke1:Center > Modify With > Tracker Position.
- 5 Click the Modifiers tab to view the Tracker controls.
- 6 From the Node Editor, drag the MediaIn for the image you painted on, and drag it into the Tracker Source field in the Inspector.



Drag the MediaIn you want to track into the Tracker Source field in the Inspector

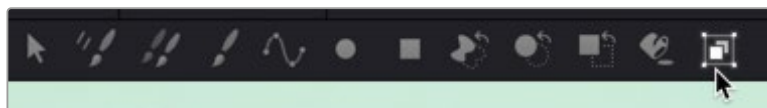
- 7 Click the Track Forward button.
- 8 After tracking, at the bottom of the Inspector, use the Tracker 1 X Offset/Y Offset controls to reposition the paint stroke, if necessary.

Tracking a Group of Paint Strokes

You can assign a tracker to multiple strokes by adding the stroke to a group and connecting the tracker to the group. Instead of connecting each individual stroke, the group's center is used for all of the strokes. Assuming the motion of each object is consistent in the same direction, as it would be objects "nailed to the set", then applying the tracker to the group makes cloning multiple objects out with a single paint node very easy.

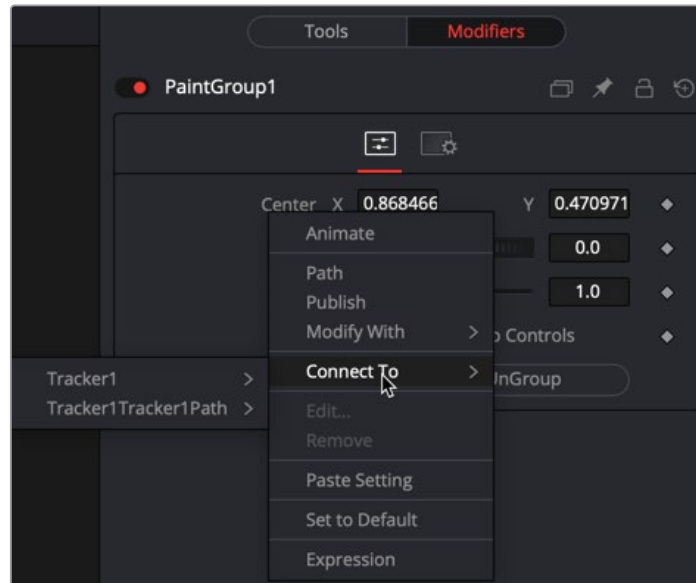
To group paint strokes, do the following:

- 1 Drag a bounding box, Shift-click, or Command-click to select every stroke that you want to group together.
- 2 Click the Paint Group button in the Paint toolbar.



Selecting all the strokes and then clicking the Paint Group button collects all the strokes into a single group

The group's onscreen controls replace the controls for each paint stroke, and the Modifiers tab in the Inspector shows the group's parameters. The individual strokes are still editable by selecting Show Subgroup Controls in the Modifiers tab of the Inspector. The group then comes with a Center, Angle, and Size control for connecting to a tracker.



Use the onscreen controls, or in the Modifiers tab, right-click over the Paint Group's Center X label to connect the tracker

Using the Planar Tracker with the Paint Tool

Here's an example that dives deeper into a workflow where we use the Paint tool with the Planar Tracker for retouching a clip. We'll eliminate some facial scars on an actor's forehead in a commercial by combining the Paint node with the PlanarTracker node, illustrating a common way of using these two powerful tools together.



The actor has some scars on his forehead that the director would like painted out

Because this is a clip in motion, we can't just paint out the scars on the man's forehead; we need to deal with the motion so that the paint work we do stays put on his face. In this case, a common workflow is to analyze the motion in the image and use it to apply a "steady" operation, pinning down the area we want to paint in place so we can paint on an unmoving surface.

Setting Up the Planar Tracker for Stabilization

The best way to do this is to use the Planar Tracker, so we'll add the PlanarTracker node after the Medialn1 node, such that the image connects to the background input of the PlanarTracker node. As always, it's important to be careful about which input you connect for the effect to work properly.



Adding a PlanarTracker node to analyze and steady the part of the image we want to paint on

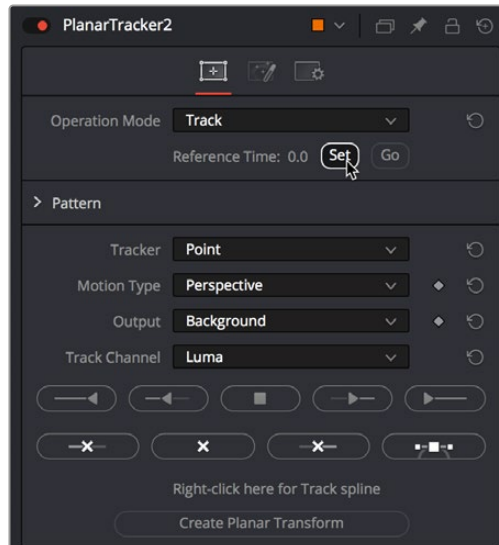
With the PlanarTracker node selected and loaded in the viewer, a viewer toolbar appears with a variety of tools for drawing shapes and manipulating tracking data. The Planar Tracker works by tracking flat surfaces that you define by drawing a shape around the feature you want to track. When you first create a PlanarTracker node, you can immediately begin drawing a shape, so in this case, we draw a simple polygon over the man's forehead since that's the feature we want to steady in preparation for painting.

We draw a simple box by clicking once each on each corner of the man's forehead to create control points, and then clicking the first one we created to close the shape.



Drawing a shape over the man's forehead to prepare for Planar Tracking

In the Inspector, the PlanarTracker node has tracking transport controls that are similar to those of the Tracker. However, there are two buttons, Set and Go, underneath the Operation Mode menu, which defaults to Track, since that's the first thing we need to do. The Set button lets you choose which frame to use as the "reference frame" for tracking, so you click the Set button first before clicking the Track Forward button below.



Setting a reference frame at the beginning of the range of frames we want to track

TIP: The Set button lets you supervise a Planar Track in progress and stop it if you see it slipping, making adjustments as necessary before clicking Set at the new frame to set a new reference before continuing to track forward towards the end of the clip.

The Pattern controls let you set up how you want to handle the analysis. Of these controls, the Motion Type menu is perhaps the most important. In this particular case, Perspective tracking is the analysis we want. Still, in other situations, you may find you get better results with the Translation, Translation/Rotation, and Translation/Rotation/Scale options.

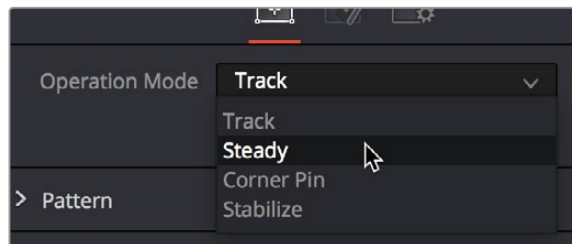
Once you initiate the track, a series of dots appears within the track region shape you created to indicate trackable pixels found. A green progress bar at the bottom of the Timeline ruler lets you see how much of the shot is remaining to track.



Clicking the Track from First Frame button to set the Planar Track in progress; green dots on the image and a green progress bar let you know the track is happening

NOTE: If nothing happens when you track, or it starts to track and then stops, that's your cue that there isn't enough trackable detail within the shape you've drawn for the Planar Tracker to work, and your best bet is to choose a different location of the image to track.

Once the track is complete, you can set the Operation Mode of the PlanarTracker node's controls in the Inspector to Steady.



Setting the PlanarTracker node to Steady

You'll immediately see the image warped as much as is necessary to pin the tracked region in place for whatever operation you want to perform. If you scrub through the clip, you should see that the image dynamically cornerpin-warps as much as is necessary to keep the forehead region within the shape you drew pinned in place. In this case, this sets up the man's head as a canvas for paint.



Steadying the image results in warping as the forehead is pinned in place for painting

At this point, you're ready to paint out those scars.

Connecting the Paint Node

Although you could paint directly on the image by connecting the Paint node after the Planar Tracker, it gives you more control over the process if you merge the Paint node over the top of the steadied image. To do that, you'll add a Merge after the Planar Tracker and then connect a Background node into the foreground of the Merge. Insert a Paint node after the background, and you are ready for this process.



Paint node merged over the top of a steadied Planar Tracker

Making the Background Node Transparent

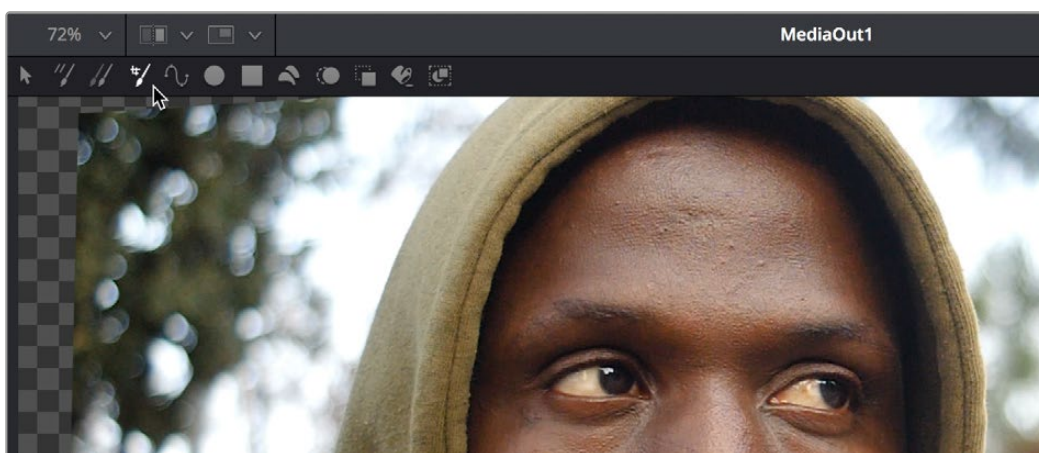
The Background node must be fully transparent; otherwise, you are painting on the solid color instead of the Merge background image. To make the background transparent, drag the Alpha channel sidebar all the way up in the Color Picker.



The Color Picker in the Background node set to be transparent

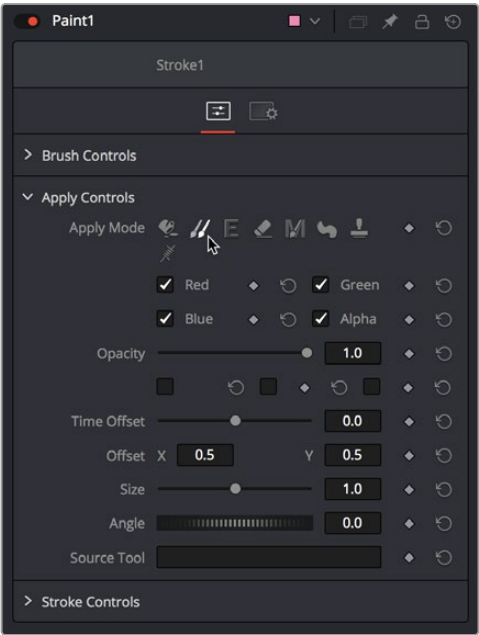
Selecting the Stroke and Clone Mode

With the Paint node selected and set up, the next thing we want to do is to select the Stroke tool. The Stroke tool is the tool of choice when you want to paint out features or paint in fixes to subjects within the frame that need to remain in place for the entire shot.



Choosing the Stroke tool from the Paint node's tools in the viewer toolbar

Next, choose the Clone mode from the Apply Controls. In this example, we'll clone part of the man's face over the scars to get rid of them. Choosing the Clone mode switches the controls of the Paint node to those used for cloning.



Choosing the Clone mode in the Inspector

Setting a Clone Source

The Paint node clones from the connected Input image unless you instruct it otherwise. If you are just painting a color over the background image, then you are good to go. However, if you plan on using the merge's steadied background image for cloning with the Paint tool, you must set that image as the Paint node's source. To set the clone source for the Paint node, you drag the PlanarTracker node into the Source Tool field in the Inspector.



Any node can be dragged into the Source Tool field when cloning with the Paint tool

With the Stroke tool selected in the Paint toolbar, the Clone mode selected in the Inspector controls, and the Source for cloning added to the Source Tool field, we're ready to start painting. If we move the pointer over the viewer, a circle shows us the paint tool, ready to go.

To use the clone brush, first hold down the Option key and click somewhere on the image to identify the source area of the clone. In this example, we'll sample from just below the first scar we want to paint. After Option-clicking to sample the image, you can click to begin painting anywhere in the frame.



Setting an offset to sample for cloning (left), and dragging to draw a clone stroke (right)

If you don't like the stroke you've created, you can undo with Command-Z and try again. We repeat the process with the other scar on the man's forehead, possibly adding a few other small strokes to make sure there are no noticeable edges, and in a few seconds, we've taken care of the issue.



Original image (left), and after painting out two scars on the man's forehead with the Stroke tool set to Clone

TIP: You can adjust the size of the brush right in the viewer, if necessary, by holding down the Command key and dragging the pointer left and right. You'll see the brush outline change size as you do this.

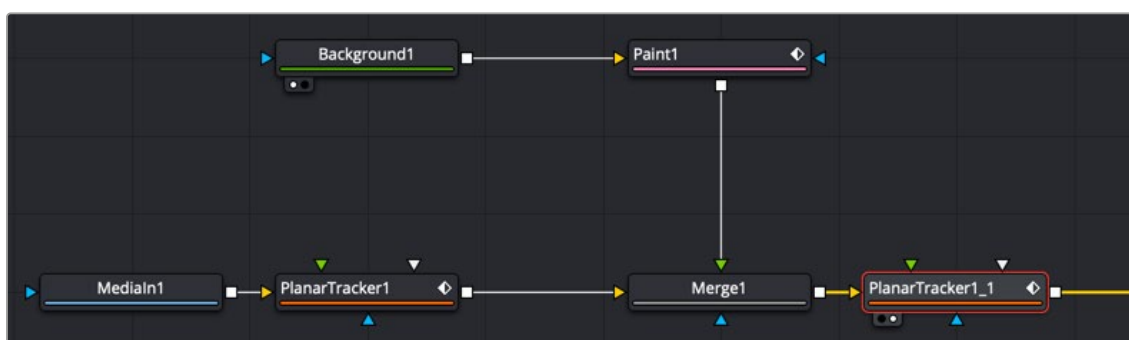
Inverting the Steady Effect to Put the Motion Back In

At this point, scrubbing through the clip shows that the paint strokes we've made are indeed sticking to the man's forehead as we need them to do. Now we just have to invert the transform that the Planar Tracker applied to restore the clip back to the way it was, except now with the painted fix attached in the process. This ends up being a two-part process, but the first part is the simplest.



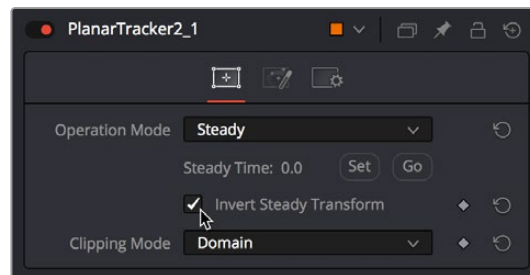
Scrubbing through the steadied clip shows the paint fix is “sticking” to the man’s forehead

We select and copy the PlanarTracker node coming before the Merge node, and paste a copy of it after. This copy has all the analysis and tracking data of the original PlanarTracker node.



Pasting a second copy of the PlanarTracker node after the Paint node

With the second PlanarTracker node selected, we go into the Inspector and turn on the Invert Steady Transform checkbox, which inverts the steady warp transform to restore the image back to the way it was.



Turning on Invert Steady Transform to try to restore the image back to the way it was

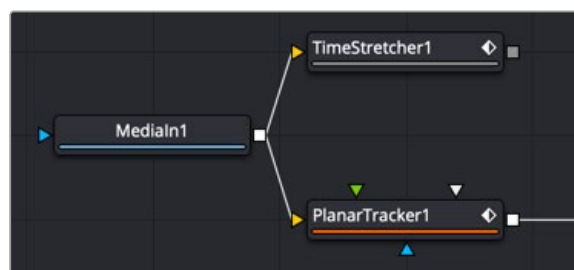
This is just one example of how to set up a Planar Tracker and Paint node. In some instances, you made need to do more work with masks and layering, but the above example gives you a good starting point.

Painting a Clean Plate

On simple clips, planar tracking the clone paint strokes may work fine. In other cases, you may not be able to steady the clip, or the strokes might appear like they are “bubbling.” Paint is just a single stroke repeated over multiple frames. Each painted frame has the potential to appear differently than the one before or after. Often, the more reliable way to use paint is by creating what is referred to as a “clean plate.” Create one good painted frame and freeze it. Then use the Planar Tracker to track the cleaned area over the top of the original. The benefit is that there is only a single paint stroke instead of repeated paint strokes that potentially show unwanted artifacts. For this next example, we continue with the previous clip, but now the technique is to freeze, mask, and composite just a single frame of the fixed forehead over the original clip.

Creating a Freeze Frame and Clean Plate

As in the previous example, you still track the clip using the Planar Tracker. After tracking, you can branch out from the MediaIn (or Loader node if you are using Fusion Studio) into a Time Stretcher to create the freeze frame.



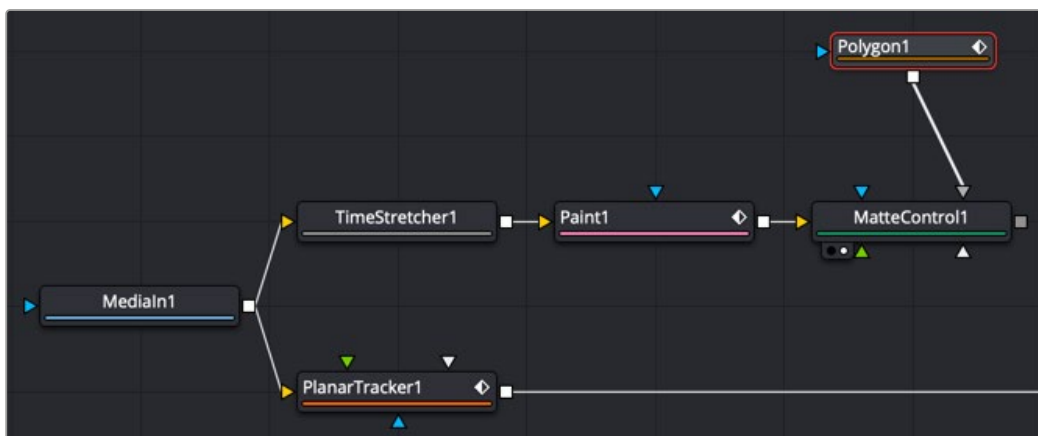
Time Remap is used to retime or freeze a clip

Disable the default Keyframe in the Time Stretcher and enter the frame you want to freeze. If you have already performed a Planar Track, then entering the frame you set as the Reference Frame is usually a good frame to freeze.

To create the clean plate, you connect the paint node to the output of the Time Stretcher. Clone over the areas you want to hide, and you now have a single clean frame. Now you need to composite the clean area over the original.

Isolating the Painted Forehead

First, we need to mask out just the man's painted forehead. We can do this by connecting a MatteControl node to the output of the Paint node and then connect a Polygon node to the garbage matte input. This lets us draw a shape with the Polygon node and use it as a mask to crop out the man's painted forehead.



Add a MatteControl node with a garbage mask to cut out the painted forehead

TIP: When it comes to using masks to create transparency, there are a variety of ways to connect one—for example, (a) attach the image to the background input of a Brightness/Contrast node and attach a Polygon mask node to the effect mask input. On the Brightness/Contrast node, enable the Alpha channel and lower the Gain slider to darken a hole, or (b) using Channel Booleans to copy channel data to the alpha from a Polygon node attached to the foreground input and the image attached to the background input.

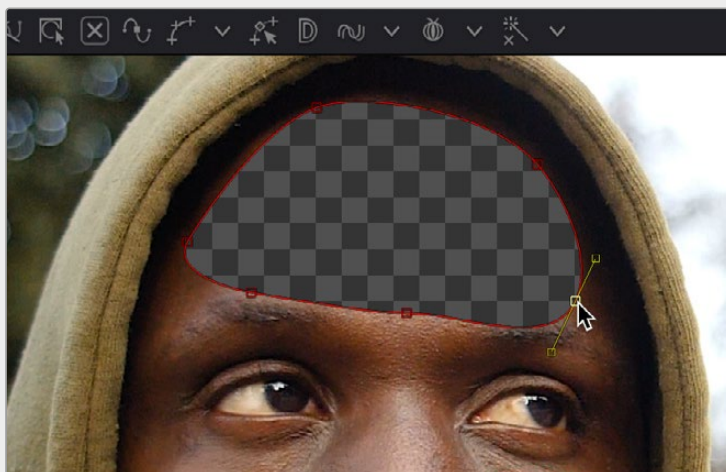
Drawing a Polygon Mask

After moving the playhead to the first frame of the clip, we're ready to draw a mask to isolate the "clean plate" forehead. Loading the MatteControl1 into the viewer and selecting the Polygon1 node so that we see its tools in the viewer toolbar sets us up for drawing a polygon.

Drawing shapes using the Polygon node is similar to shape drawing in other spline-based environments, including the Color page:

- Clicking once draws a corner control point.
- Clicking and dragging creates a Bézier curve.
- Clicking the first control point you created closes a shape.

We click and drag to create a shape that outlines the man's forehead, and when we close the shape, we see exactly the opposite of what we want, a hole in the middle of the image.

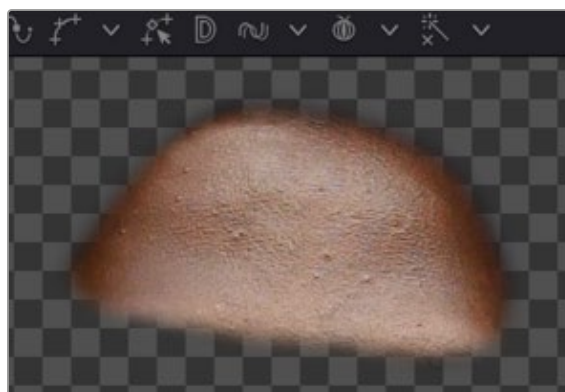
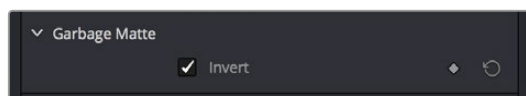


Drawing a shape to isolate the forehead gives an inverted result at first when using the Garbage Matte input of the MatteControl node to attach the Polygon to the MatteControl node

Before fixing this, we drag the Soft Edge slider in the Inspector to the right to blur the edges just a bit.

Inverting the Garbage Input

Selecting the MatteControl1 node, we open the Garbage Matte controls and click the Invert checkbox, which immediately gives us the result we want, of the forehead in isolation, ready for compositing.



Inverting the Garbage Matte input (Left), and the resulting inverted mask inverting the forehead (right)

Compositing the Painted Forehead Against the Original Image

Almost finished. Next, we'll add a Merge node to layer the fixed forehead over the original image being output by the MediaIn node.

We create a Merge node connected to the output of the PlanarTracker node, and then we connect the MatteControl's output to the green foreground input of the Merge node. This puts the cropped and fixed forehead on top of the original image.



The painted forehead composited against the original image

Match Moving the Mask to the Shot

Now we have the best of both worlds: a fixed forehead and the background of the shot looking good. However, if we select the Polygon node and then scrub forward in the clip, the fixed forehead mask drifts out of sync with the motion of the shot, so we have one remaining issue. Fortunately, match moving the mask to move with the shot is really simple.



Because the Polygon isn't animated to match the motion of the shot, it goes out of sync

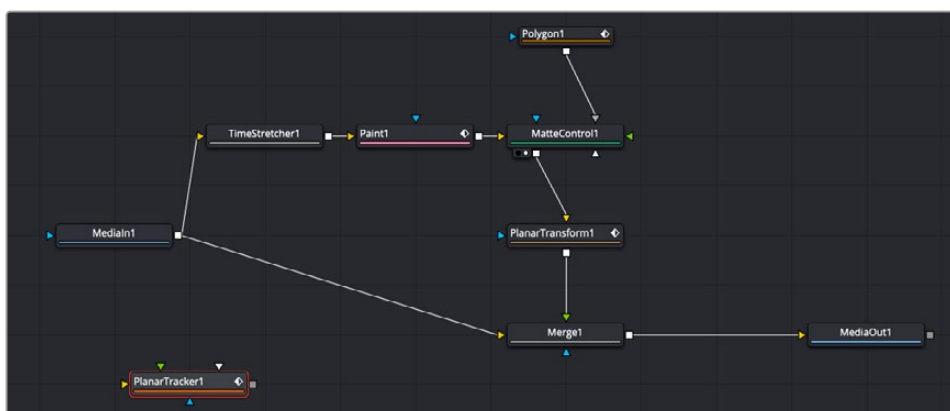
Selecting the first PlanarTracker node that comes right after the MediaIn node, and choosing Track from the Operation Mode menu, reveals a Create Planar Transform button at the bottom of the listed controls. Clicking this button creates a new, disconnected Planar Transform node in the Node Editor, which has the transforms from the Planar Tracker baked in. Unlike the Planar Tracker, the Planar Transform is intended for use with masks. With the Planar Transform created, there is no longer a use for the Planar Tracker, and it can be disconnected or deleted from the Node Editor.

We can insert this new node into the node tree to use it by holding down the Shift key and dragging the node over the connection between the Polygon node and the MatteControl node, dropping it when the connection highlights.



Inserting a PlanarTransform node by holding down the Shift key while dropping over a connection (left), and after inserting the PlanarTransform node (right)

With the new Planar Transform node inserted, the Polygon automatically moves to match the motion of the forehead that was tracked by the original PlanarTracker node, and it animates to follow along with the movement of the shot. At this point, we're finished!



The final painted image, along with the final node tree